📘 AI ARCHITECTURE DOCUMENTATION

Product: ShipGen – Autonomous Logistics Operator

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1. 🎯 OBJECTIVE

Define:

- Where AI is used in the system
- What type of AI models are used
- How each AI component functions
- How AI enables automation across the entire logistics flow

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2. 🧠 AI SYSTEM OVERVIEW

ShipGen uses **multi-layer AI architecture**:

1. Understanding Layer (NLP)
2. Decision Layer (Assignment & Planning)
3. Optimization Layer (Routing & Resource usage)
4. Prediction Layer (Delay & anomaly detection)
5. Action Layer (Auto-resolution & communication)

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3. 🔄 END-TO-END AI FLOW

Order Input → NLP → Trip Decision → Assignment Optimization → Tracking → Prediction → Auto Resolution → Learning

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4. 🧩 MODULE-WISE AI IMPLEMENTATION

4.1 ORDER INGESTION AI

🎯 **Purpose:**

Convert unstructured input → structured data

🧠 **AI Type:**

- NLP (Natural Language Processing)
- LLM (Large Language Model)

⚙️ **Functionality:**

- Parse messages (Email / WhatsApp)
- Extract:
  - Pickup location
  - Drop location
  - Load details
  - Time

📌 **Example:**

Input:
"Send 10 tons from Vizag to Hyderabad tomorrow"

Output:
{
pickup: Vizag
drop: Hyderabad
load: 10 tons
date: tomorrow
}

🛠️ **Suggested Models:**

- OpenAI GPT (initial)
- Fine-tuned domain NLP model (future)

4.2 TRIP DECISION ENGINE

🎯 **Purpose:**

Automatically create trip

🧠 **AI Type:**

- Rule-based AI + ML hybrid

⚙️ **Functionality:**

- Validate order
- Generate trip structure
- Prepare assignment candidates

Logic:

- If data complete → auto-create trip
- If incomplete → request clarification

🛠️ **Models:**

- Decision Trees
- Rule Engine (initial phase)

4.3 DRIVER & VEHICLE ASSIGNMENT AI

🎯 **Purpose:**

Select best driver and vehicle

🧠 **AI Type:**

- Optimization Algorithm
- Scoring Model (ML-based)

⚙️ **Functionality:**

Inputs:

- Driver availability
- Distance to pickup
- Driver rating
- Vehicle capacity

Scoring Logic:

Score = (distance + performance + cost efficiency + availability)

Output:

- Best match (Top 1 / Top 3)

🛠️ **Models:**

- Linear scoring model (initial)
- ML ranking model (future)

4.4 ROUTE OPTIMIZATION AI

🎯 **Purpose:**

Select best route

🧠 **AI Type:**

- Optimization algorithms

- Heuristic + ML hybrid

⚙️ **Functionality:**

- Shortest path
- Traffic-aware route
- Toll cost optimization

Output:

- Primary route
- Alternate routes

🛠️ **Tools:**

- Google Maps API / Mapbox
- Custom optimization logic

4.5 TRACKING & MONITORING AI

🎯 **Purpose:**

Monitor trip in real-time

🧠 **AI Type:**

- Anomaly Detection
- Pattern Recognition

⚙️ **Functionality:**

- Detect:
    - Unusual stops
    - Route deviation

    - Inactivity

Output:

- Events generated

🛠️ **Models:**

- Rule-based detection (initial)
- ML anomaly detection (future)

4.6 PREDICTION ENGINE

🎯 **Purpose:**

Predict delays before they happen

🧠 **AI Type:**

- Time-series prediction models
- Classification models

⚙️ **Functionality:**

Inputs:

- GPS data
- Route data
- Historical trips

Output:

- Delay probability
- Updated ETA

🛠️ **Models:**

- Regression models
- LSTM / time-series ML (future)

4.7 AUTO RESOLUTION AI

🎯 **Purpose:**

Solve problems automatically

🧠 **AI Type:**

- Recommendation System
- Decision AI

⚙️ **Functionality:**

If delay detected:

- Suggest alternate route
- Suggest driver reassignment

Output:

- Action recommendation
- Auto-triggered actions

🛠️ **Models:**

- Rule-based recommendations
- Reinforcement learning (future)

4.8 COMMUNICATION AI

🎯 **Purpose:**

Automate communication

🧠 **AI Type:**

- Template-based + LLM

⚙️ **Functionality:**

- Generate messages:
  - Trip confirmation
  - Delay alerts
  - Delivery updates

Output:

- Context-aware messages

🛠️ **Models:**

- LLM (text generation)

4.9 COST & PROFIT OPTIMIZATION AI

🎯 **Purpose:**

Maximize profit per trip

🧠 **AI Type:**

- Optimization models

⚙️ **Functionality:**

- Calculate cost:
  - Fuel
  - Driver
  - Toll
- Suggest:
  - Better route
  - Better assignment

Output:

- Profit insights

4.10 LEARNING SYSTEM (SELF-IMPROVEMENT)

🎯 **Purpose:**

Improve system over time

🧠 **AI Type:**

- Machine Learning (continuous learning)

⚙️ **Functionality:**

- Learn from:
  - past delays
  - driver performance
  - route efficiency

Output:

- Better future decisions

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5. 🧠 AI LAYERS SUMMARY

Layer	AI Role
Understanding	NLP
Decision	Rule + ML
Optimization	Assignment + Routing
Prediction	Delay detection
Action	Auto resolution

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6. ⚙️ AI IMPLEMENTATION STRATEGY

Phase 1 (MVP):

- Rule-based + basic ML
- LLM for NLP

Phase 2:

- ML models for prediction
- Ranking models for assignment

Phase 3:

- Reinforcement learning
- Fully autonomous system

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7. ⚠️ RISK & MITIGATION

Risk:

Incorrect AI decisions
👉 Mitigation:

- Human approval layer

Risk:

Data quality issues
👉 Mitigation:

- Validation system

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8. 🚀 FINAL AI VISION

> "ShipGen is not a single AI model — it is a coordinated system of multiple AI models working together to fully automate logistics operations."